

Database Systems Design & UML Modeling

Part A: Conceptual EERD

Part A Project Brief: Vacoliday

Vacoliday is a touristic company that offers high-end highly-customizable holidays, with a great emphasis on localness and well-being, in Vacomedita, a lovely town on the Mediterranean coast.

Vacoliday has several residences around the town of Vacomedita which they use to accommodate holiday makers. These residences offer two types of bookable accommodation: fully-integrated apartments and stand-alone rooms. The apartments are typically larger with one or more bedrooms and other rooms like kitchens, bathrooms, washing rooms, dining rooms, and lounges. The standalone rooms are essentially bedrooms with possibly an en-suite bathroom.

The company offers various attractive meal options that can also be booked. Collective meals can be taken at specific times at one of the stylish dining suites available at the Vacoliday residences. However, meals can also be delivered to people's accommodation, if pre-booked in advance.

Besides, Vacoliday offers a wide range of exciting bookable experiences. These experiences can take place on-site at a specific residence or off-site, out and about the area surrounding Vacomedita. There are three main kinds of on-site experiences: fitness classes (such as yoga, pilates or any other well-being physical activities) under the guidance of a fitness instructor; craft workshops during which Vacoliday customers can learn a local craft with a workshop instructor; and cooking classes for which holidaymakers are taught how to prepare local dishes by a chef. On the other hand, offsite experiences are led by a guide and come mainly under two broad categories: excursions to places of interest and walking hikes. Thus, hikes are on foot and involve walking in the countryside or wilderness around Vacomedita. Excursions are run by a driver on a bus and allow the visiting of places of interest, beauty spots and other landmarks across the stunning Vacomedita region.

In terms of staffing, Vacoliday relies on highly-trained employees with well-defined roles. Guides oversee all off-site experiences. Drivers use their buses to accompany people on excursions. Chefs take care of meals but also teach cooking classes. Housekeeping attendants are responsible for the good maintenance of their allocated accommodations. Finally, instructors come under two groups: the fitness instructors who run the exercise classes for well-being and the craft instructors who lead and deliver the workshops in which people learn exciting local craft skills.

Part A Questions

You have been hired by Vacoliday as a **Database Architect** to undertake a database project to support the data needs of the company. Your job in this first part is to investigate the Vacoliday company, produce a high-quality **CONCEPTUAL ENHANCED ENTITY RELATIONSHIP DIAGRAM (EERD)** to support its data needs and document it with a data dictionary.

Part B: Logical ERD

Part B Project Brief: Tourmato

Tourmato is a touristic company that offers exciting “off-the-beaten track” visiting tours around several cities across Europe. Essentially, Tourmato takes groups of customers around European cities to visit them and experience their unique atmospheres, under the direction of a local experienced tour guide. To provide a range of experiences, every city covered by Tourmato offers many visiting tours. A tour session is a visiting tour that has been assigned a specific start date and time, a start address, an end date and time, an end address and a specific price. Tourmato customers can place a booking for a variety of exciting tour sessions, depending on what they like.

The conceptual EERD for Tourmato is given in Figure 1 (see next page).

Part B Questions

You have been hired by Tourmato as a **Database Consultant** to undertake a database project to support the data needs of the firm. In this second part, you are given a conceptual data model for Tourmato (figure 1), and your first goal is to **map it** onto a high-quality **LOGICAL ENTITY RELATIONSHIP DIAGRAM (ERD)** to logically represent how the key business data needs can be organised as a set of interrelated tables that can then be implemented. These tables need to be interconnected according to the strict rules of the relational model to be implementable.

Figure 1. Conceptual EERD for Tourmato



